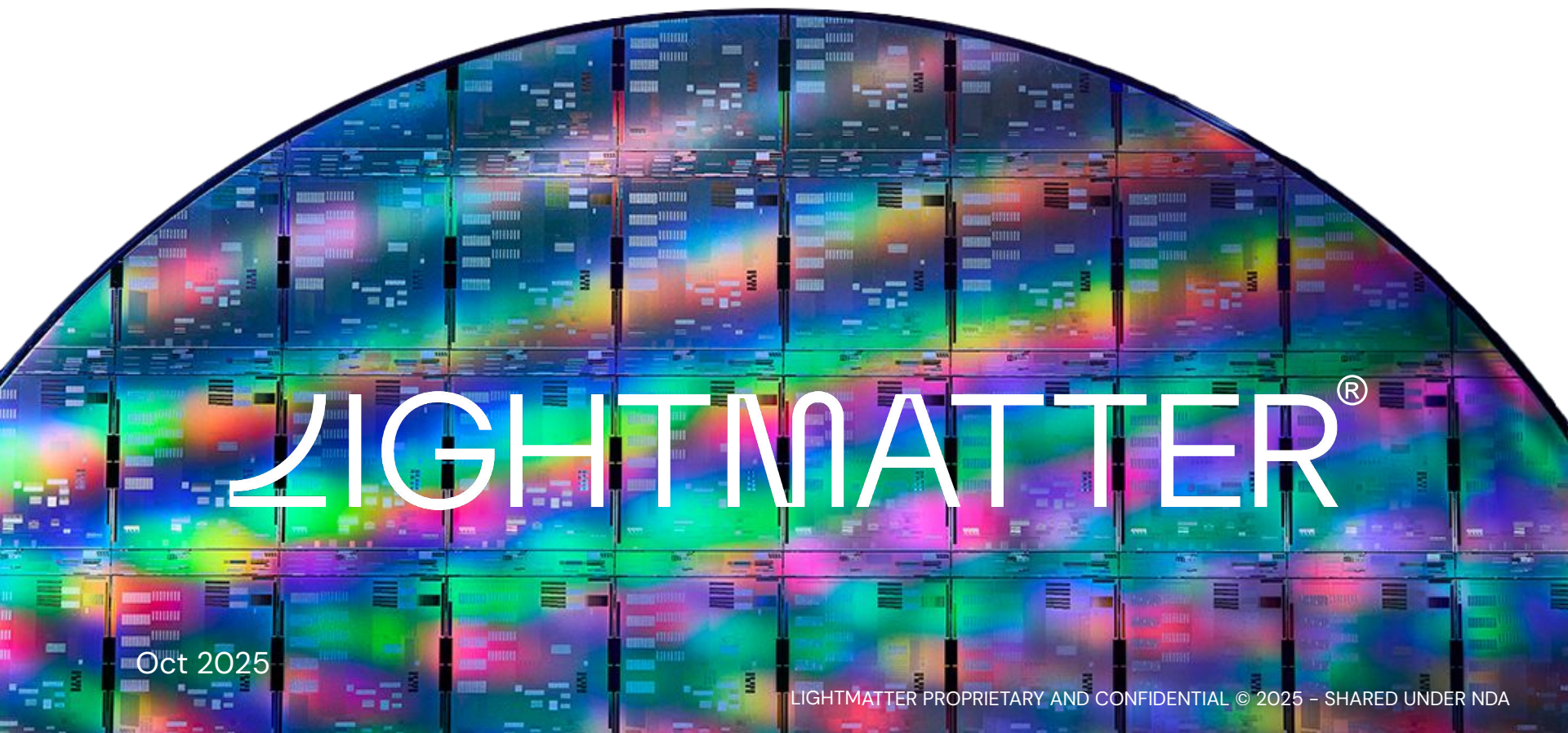


The photonic (super)computing company



LIGHTMATTER®

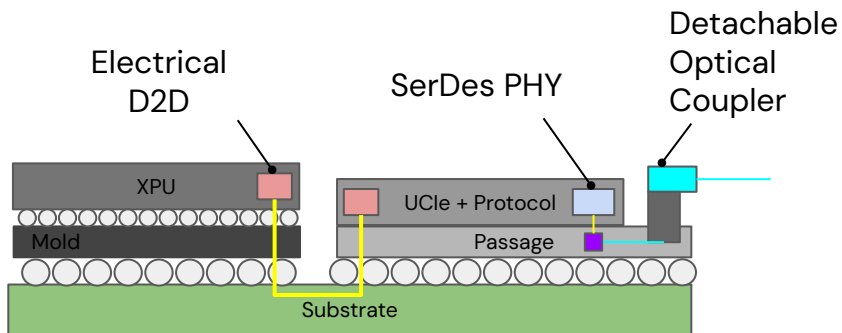
Oct 2025

Agenda

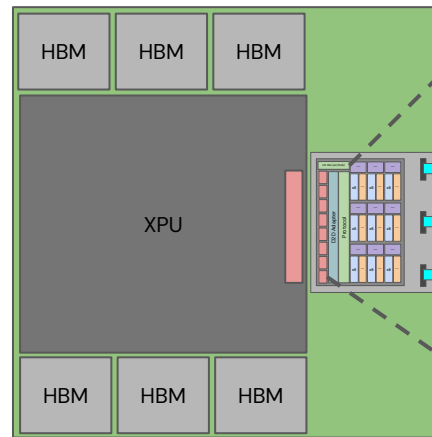
- Product architectures
- SerDes requirements – power, performance, area
- Co-optimization opportunities
- Schedule and Plan

Product Concept: XPU + 3D CPO

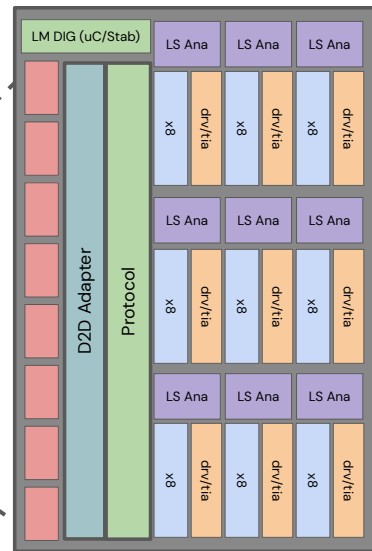
- Up to 28.8T per XPU edge with 2x OEs
- OE integration on Substrate today, CoWoS in future with support for AP UCle D2D PHYs
- 200G DR or 53Gx4 with options for BiDi & DWDM link architectures
- Interoperable with VHP ELSFPs & Guide for DWDM



XPU + OE Cross-Section



XPU



EIC Layout

OCI MSA Specifications Overview

OCI PMA + PMD Specifications

- OCI leverages 200GBASE-R PCS + PMA
- Specifies a OCI PMA that implements the MUX and deskew (implementations from 200G up to 1.6TBASE-R defined)
- 1st Generation OCI-PMD using 53.125Gbps NRZ SerDes (4x lanes)

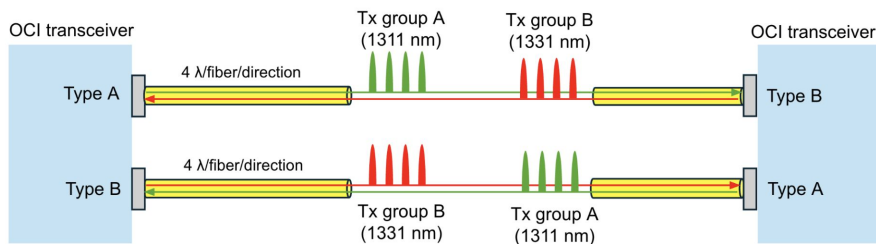
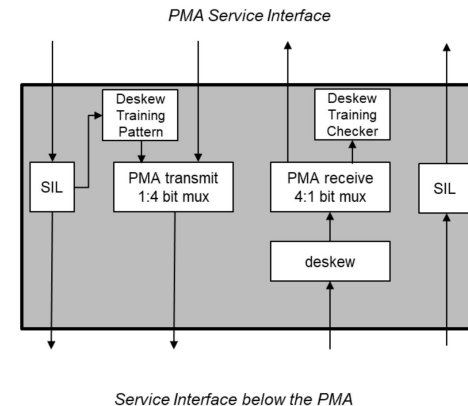
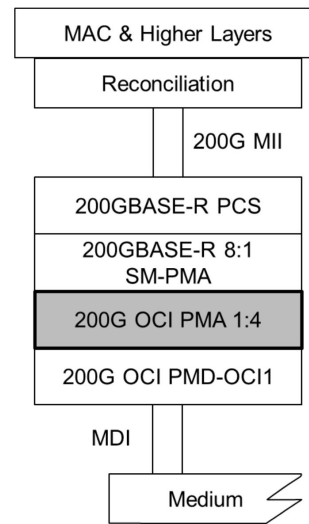


Figure 2-1. Illustration of DWDM wavelength channels and CWDM wavelength groups for a bidirectional link.

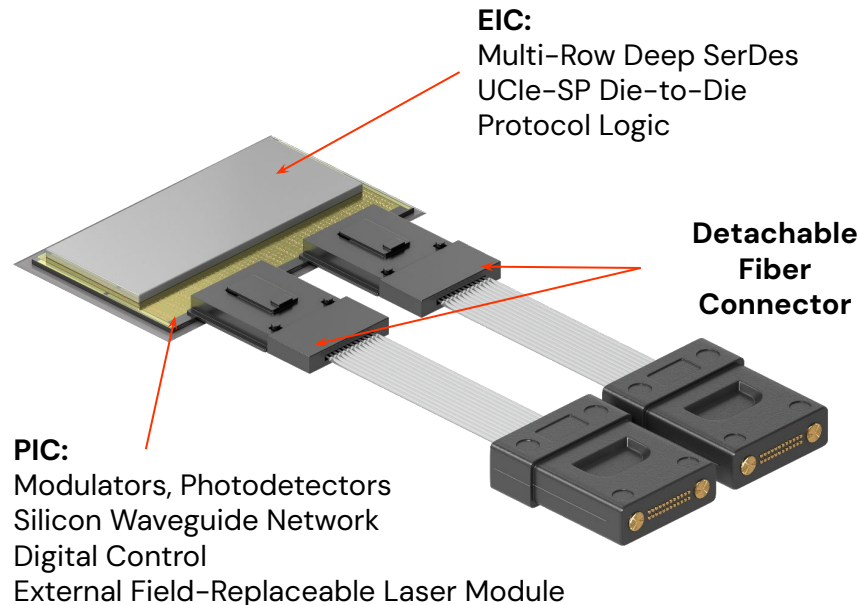
OCI Wavelength Grid Specifications

- Group A: 4x wavelengths centered around 1311 nm with 400 Ghz spacing
- Group B: 4x wavelengths centered around 1331 nm with 400 Ghz spacing

Product Concept: 3D CPO OE

OCI MSA Compliant

- Passage L-Series CPO
 - 3D Optical engine with industry aligned DWDM link architecture
 - UCle-S 64G D2D Host Interface
- Timeline
 - ES Q4'27
- Goals
 - Demonstrate 3D OE product design, link-level reliability, and package integration with XPU silicon
 - Pipe clean supply chain and manufacturing flow for HVM of L-Series 3D OE
- Additional Benefits
 - EVK available with telemetry, PRBS pattern generator, and full network traffic insertion

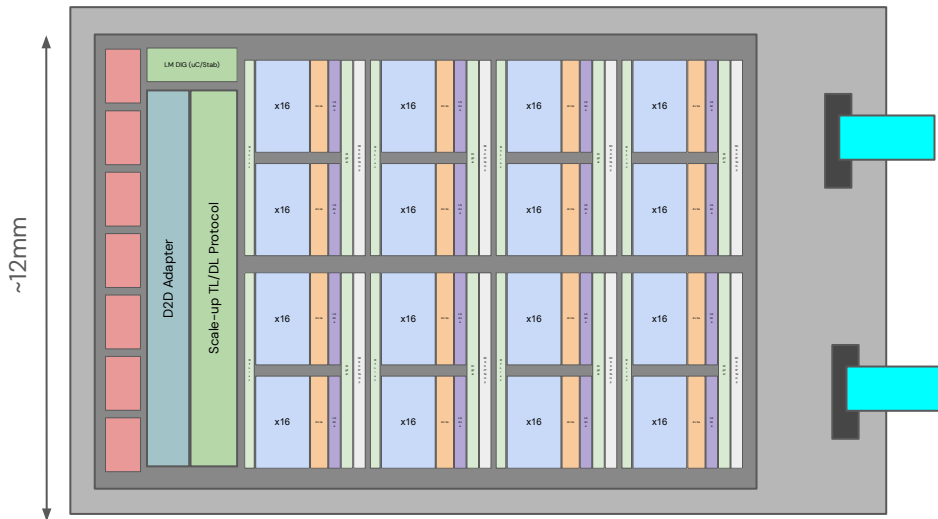
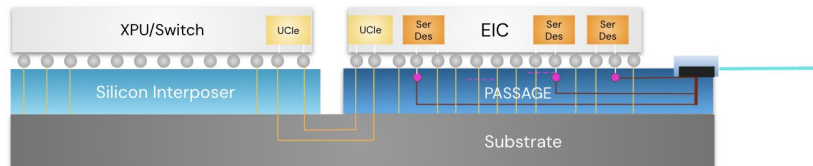


Product Concept: 3D OE CPO

OCI MSA Compliant

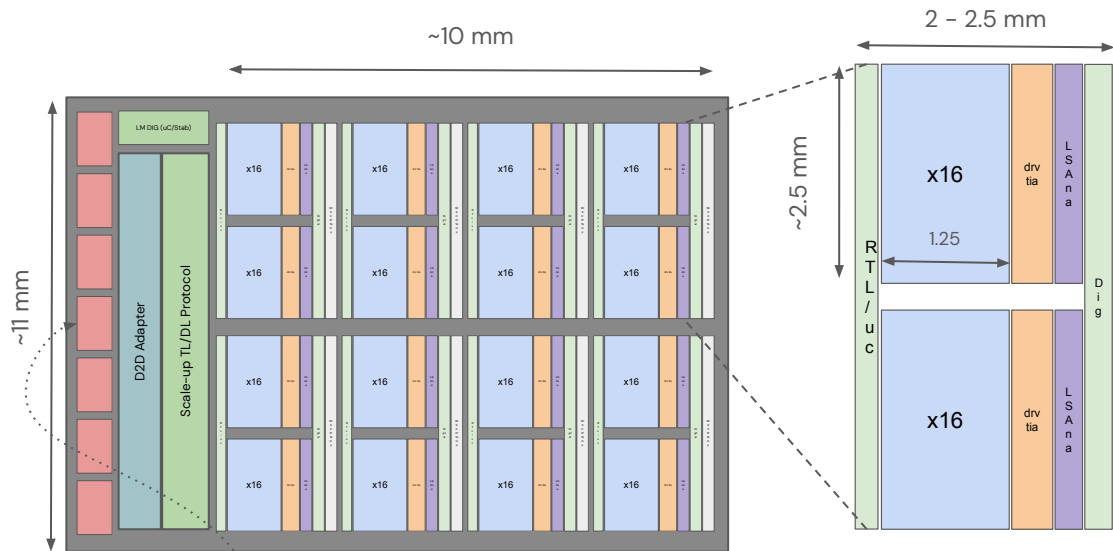
Specifications

- OE Bandwidth: 12.8Tbps Tx + 12.8Tbps Rx
- Host Interface: UCle-SP 64G
- SerDes: 56G/64G NRZ x16 Macro
- Link Architecture: 4 + 4 DWDM BiDi
- Ports: 64
- Detachable FAU
- Protocol Support: UAL, ESUN, Stream and Other



Bandwidth Density: ~2 Tbps/mm (Tx + Rx)
Estimated Power Consumption: 3pJ/bit (Inc. Laser)

Sizing: OCI MSA



UCle at 2Tbps/1.6mm
shoreline sets height to
~11mm

SerDes Lane area ~
0.15 - 0.2 mm²

Not including non-SerDes blocks
DRV/TIA are at 0.04mm² each
(0.3x0.15) - with opportunities to
shrink via integration

Product Concepts and Key PPA Metrics

Electrical Standards Compliant DR	OCI MSA DWDM
--------------------------------------	-----------------

	DR 200G	DR 400G	Gen 1: DWDM 200G	Gen 2: DWDM 400G
Line Rate [Gbps]	212.5	425	53.125 - 64	106.25 - 128
Modulation	PAM4	PAM4	NRZ	NRZ
Port BW [Gbps]	200	400	200	400
Unidi/Bidi	Unidi	Unidi	Bidi	Bidi
BW/fiber [Gbps]	200	400	400	800
SerDes Power [pJ/bit] target	2	2	1-2	1-2
PIC Power [pJ/bit] target	2	2	1	1
Laser Power [pJ/bit] target	1	1	1	1
Area/lane [mm^2] target	0.5	?	0.15-0.2	0.15-0.2
Max BER	1.00E-06	1.00E-06	1.00E-12	1.00E-12
Max symbol errors/RS FEC frame	10	10	3	3
Latency	>100ns	>100ns	20-50ns	20-50ns

SerDes Optimization Opportunities for Optics

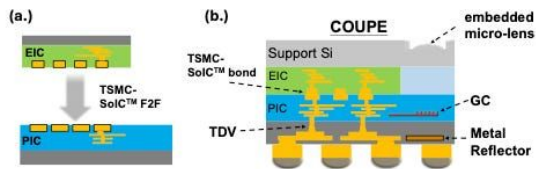
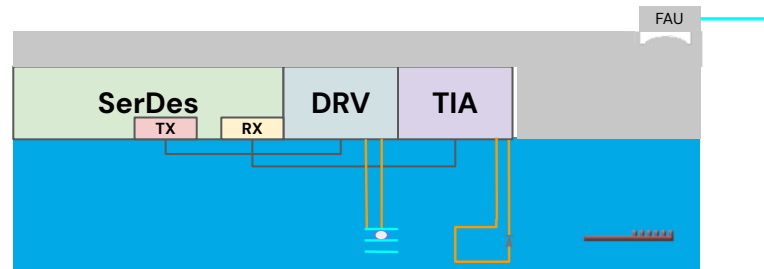
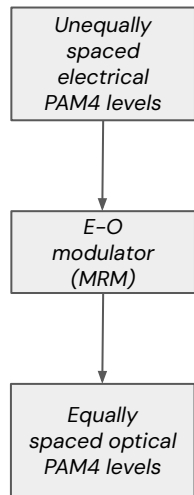


Fig. 2 (a) TSMC-SolC™ face-to-face (F2F) technology for EIC and PIC bonding. (b) COUPE critical components consist of TSMC-SolC™ bond, TDV, embedded micro-lens, and metal reflector.

COUPE Integration (Hybrid Bond)

Optical Nonlinearity Compensation



Driver/TIA integration within the SerDes IP

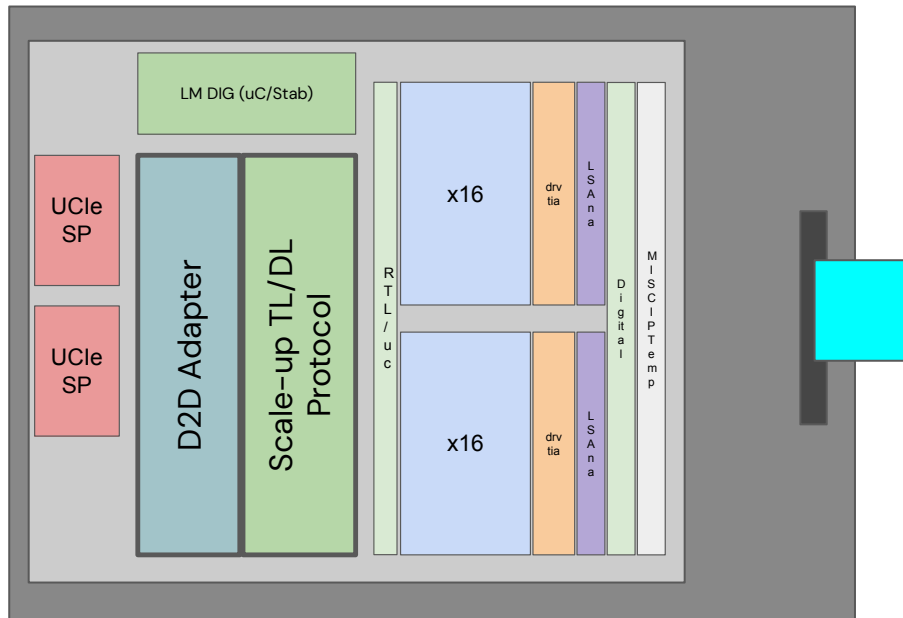
Additional opportunities

- Power optimization for standards compliant 200G DR/FR4 links
- Power and area optimization for 50G – 64G NRZ DWDM links
- Dual NRZ Drive for segmented modulators
- 400G Optical SerDes

More Details - SerDes Optimization Areas

Feature / Requirement	NRZ (OCI Optimized)	PAM4 (Standard 224G)
Tx Driver Output Swing	Programmability a Must	Programmability a Must
RX TIA AGC	Dynamic	Dynamic
FEC Choice	Bypass / Light CRC / Lite FEC	RS(544, 514)
Equalization Chain	Linear Only	Full Stack: TX FIR + LUT + CTLE + Multi-tap FFE/DFE + MLSD.
Adaptive Algo	Continuous	Continuous
Squelch / SD / Data Valid	Manual	Manual
Slicer Thresholds	Fixed Center	Independent/Asymmetric
CDR Phase Offset	N/A	Centered: Optimized for maximum horizontal eye opening.

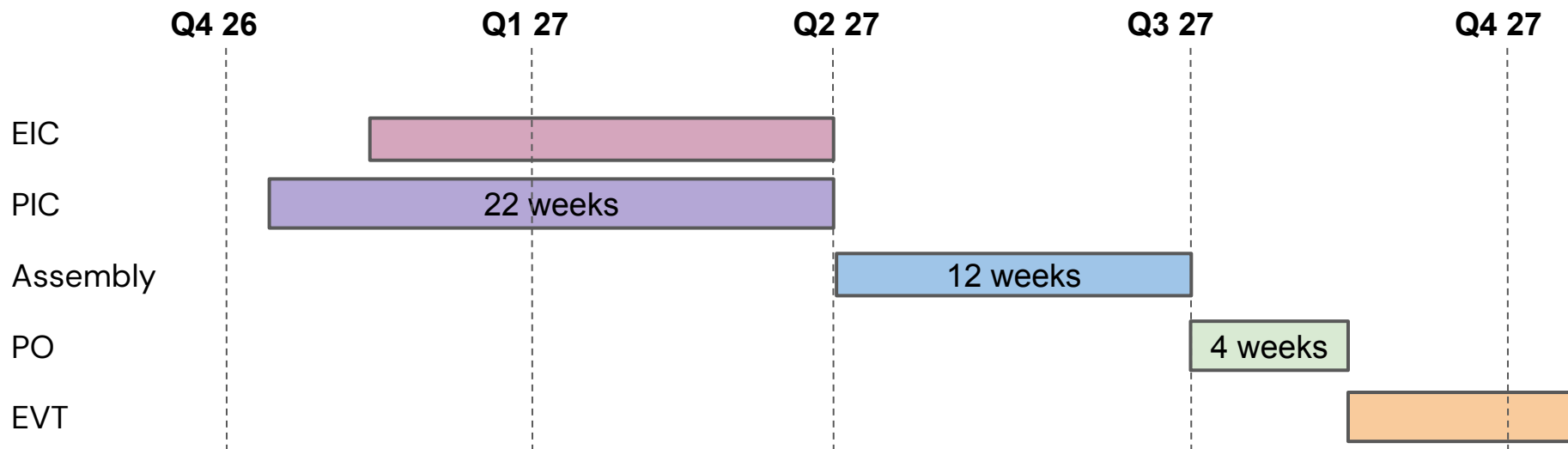
Test Chip Proposal



Features

- 4 to 8 Tbps of UCle ingress BW
- 3.2 to 6.4 Tbps of optical BW
 - 32x to 64x 106G NRZ OCI Gen2 MSA (Also demonstrates OCI Gen1 MSA)
 - 16x to 32x 212G PAM4 DR
- Available path to direct productization with customizable BW using multiple PICs

High Level Schedule



THANK YOU

XPU + 3D CPO

UCle + DR Standards Compliant

14.4Tbps 3D OE

- 200G DR Unidirectional Link Architecture (BiDi & DWDM also possible)
- Direct integration of UCle + scale up protocol die on PIC via HBI
- Detachable FAU
- UCle 64Gbps D2D for host-interface
- 224G PHY & Protocol Logic for optimal interoperability

Option 1: 2X 7.2Tbps PIC Configuration

72 Tx + 72 Rx Fibers + 18 Laser (1:4 split)

Est. EIC Dimensions: 10mmx9mm

Option 2: Single 14.4Tbps PIC Config.

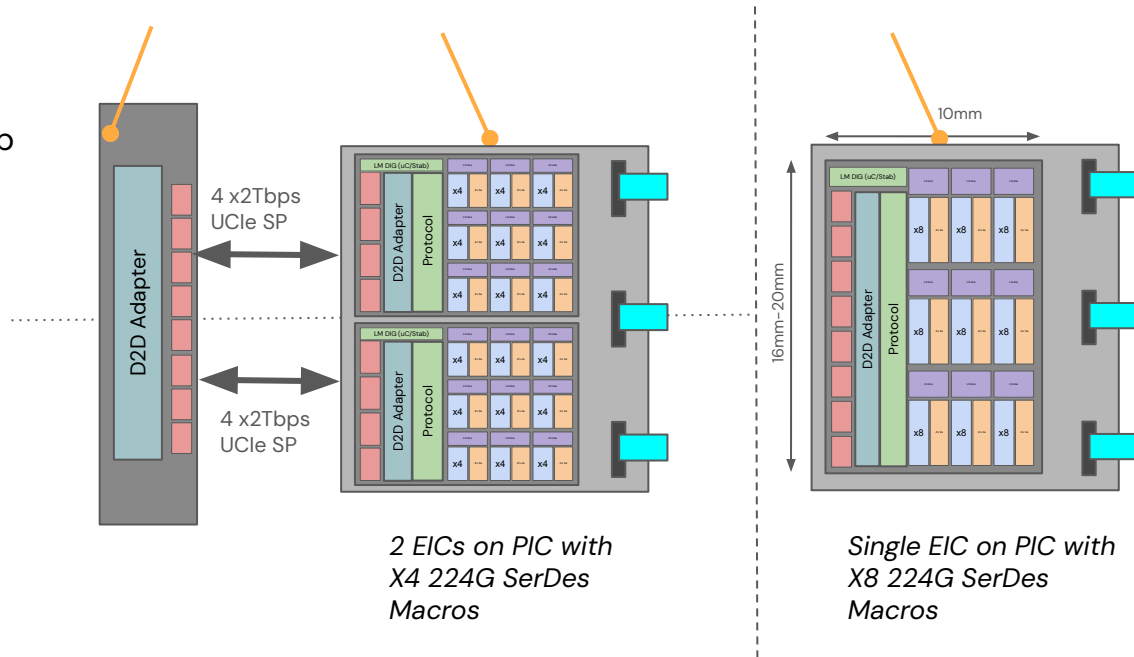
36 Tx + 36 Rx Fibers + 9 Laser (1:4 split)

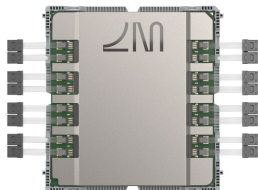
Est. EIC Dimensions 10mm x 18mm

Custom GPU ASIC

3D Stacked PIC

3D Stacked PIC



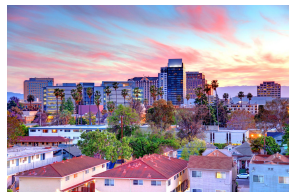


\$850M RAISED

Fidelity
Google Ventures
Sequoia
Spark Capital
Viking

T. Rowe Price
Matrix Partners
SIP Capital
MIT
Stanford

3
OFFICES



Mountain View (HQ)



Boston



Toronto

300
EMPLOYEES



Nicholas Harris, PhD
Founder, CEO



Simona Jankowski
CFO



Sujatha Wagle
VP, Supply Chain Ops



Boon Tan
VP, Product Engineering



Praveen Kukkamalla
VP, Sales



Darius Bunandar, PhD
Founder, Chief Scientist



Ritesh Jain
SVP, Engineering & Ops



Beth Keil
SVP, People



Colin Sturt
General Counsel



Kurt von Hausen
VP, Cloud Services Sales



Thomas Graham
Founder



Bob Turner
SVP, Sales & Solution Arch



Steve Klinger
VP, Product

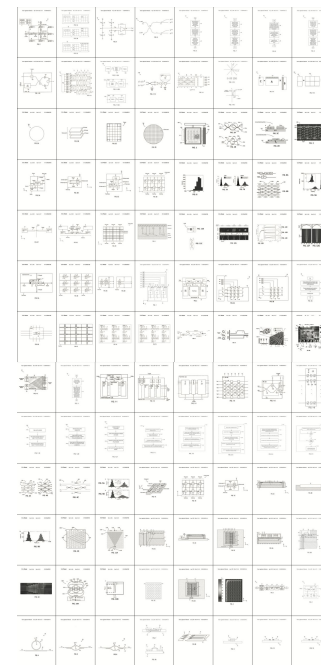


Kaushik Patel, PhD
VP, Photonics & Si Eng



Israel Kandarian
Head of Creative Marketing

311
PATENTS



Granted & Pending

LIGHTMATTER

250 Bryant Street



Our labs



MOUNTAIN VIEW
CUSTOMER DEVICE BRING-UP



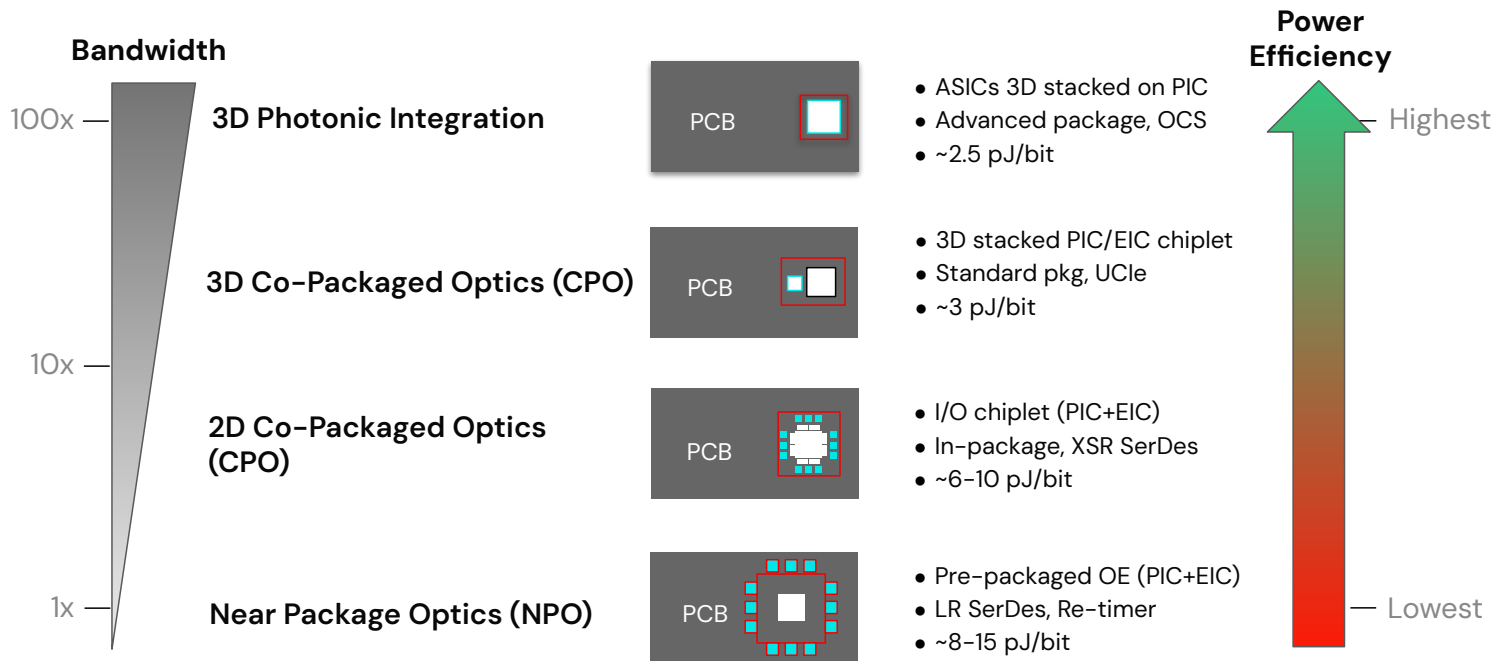
PALO ALTO
LINK-LEVEL TESTING AND VALIDATION



BOSTON
PHOTONIC DEVICE-LEVEL CHARACTERIZATION

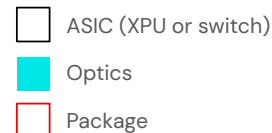


Photonic Engines Moving Closer to the ASIC

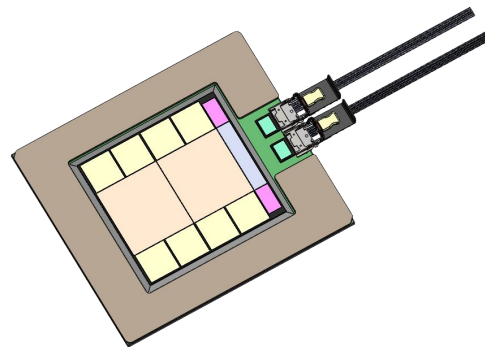
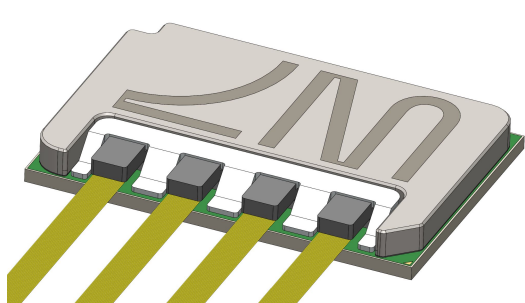


Photonic Integrated Circuit: PIC
Electronic Integrated Circuit: EIC
Optical Engine (OE) = PIC+EIC

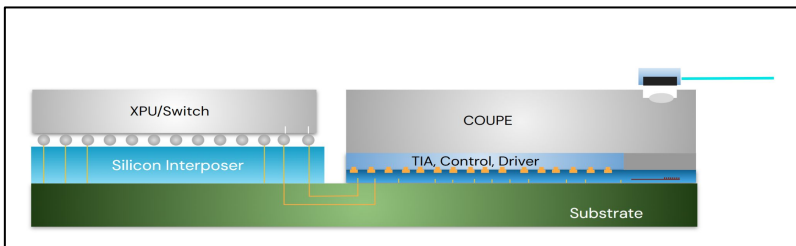
Copper, Pluggables



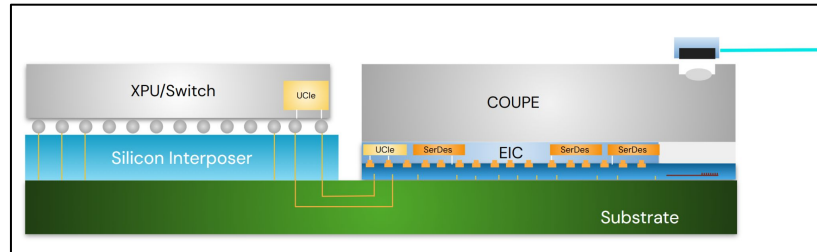
Passage L-Series



L-20 (2D Integration) On/Off Package

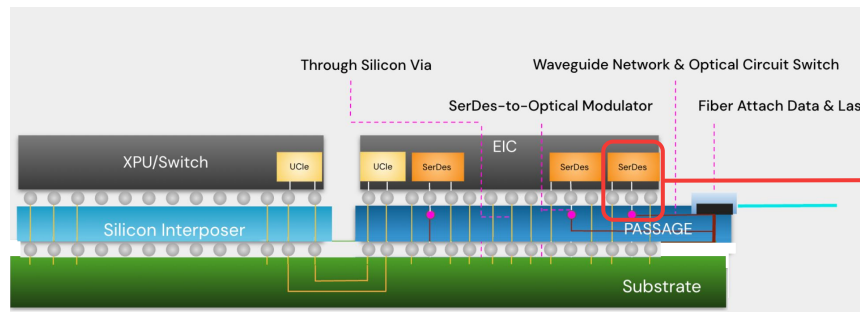


L-2xx (3D - I/O Chip on Passage)



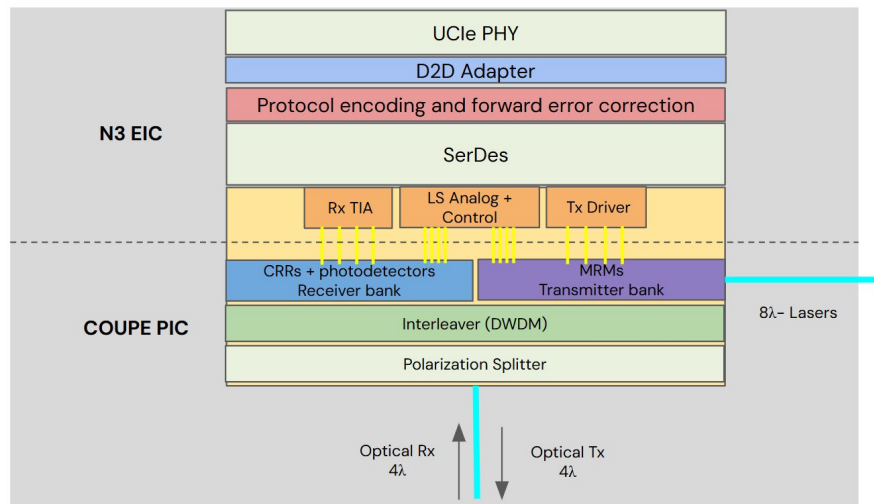
PASSAGE™

3D Integration



Cross Section

COUPE Product



Enabling All Link Architectures

MULTIPLE ROADMAP GENERATIONS HERE TODAY

STANDARDS COMPLIANT

OR

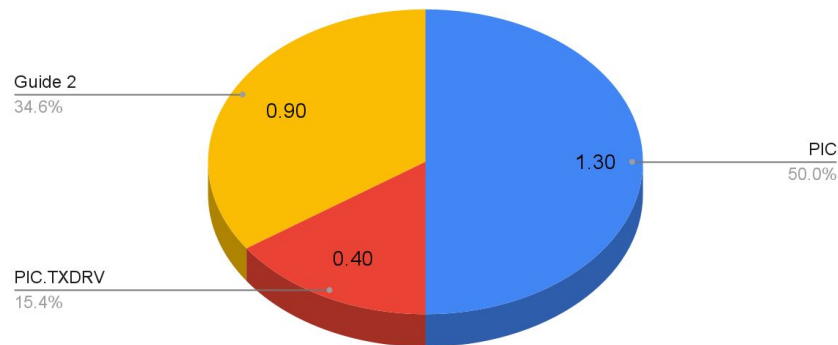
CUSTOM ADVANCED

Bandwidth	Modulation	Number of Wavelengths	Transmission Type	CWDM/DWDM
				WDM Type
56 Gbps	NRZ	16	Bi-directional	DWDM
56 Gbps	NRZ/PAM4	16	Uni-directional	DWDM
112 Gbps	PAM4	16	Uni-directional	DWDM
224 Gbps	PAM4	4	Bi-directional	CWDM
224 Gbps	PAM4	4	Uni-directional	CWDM

BOBCAT PASSAGE LINK PLATFORM

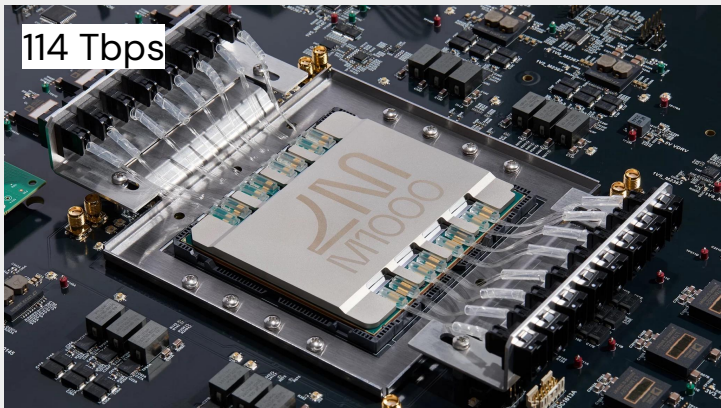
World-class energy efficiency.
Passage™ and Guide™ achieve
2.6 pJ/bit—measured in the lab.

PIC	1.3 pJ/bit
PIC.TXDRV	0.4 pJ/bit
GUIDE 2	0.9 pJ/bit



BOBCAT AND LASER

114 Tbps



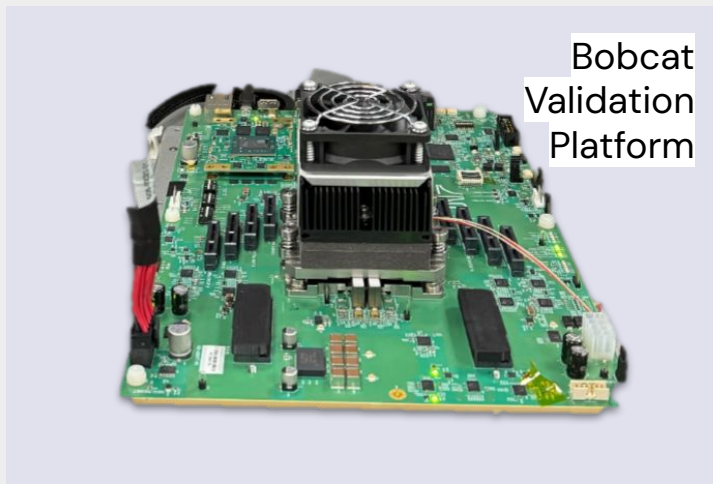
800 Gbps / Fiber



16 Lambda
1.8 Watts



Bobcat
Validation
Platform

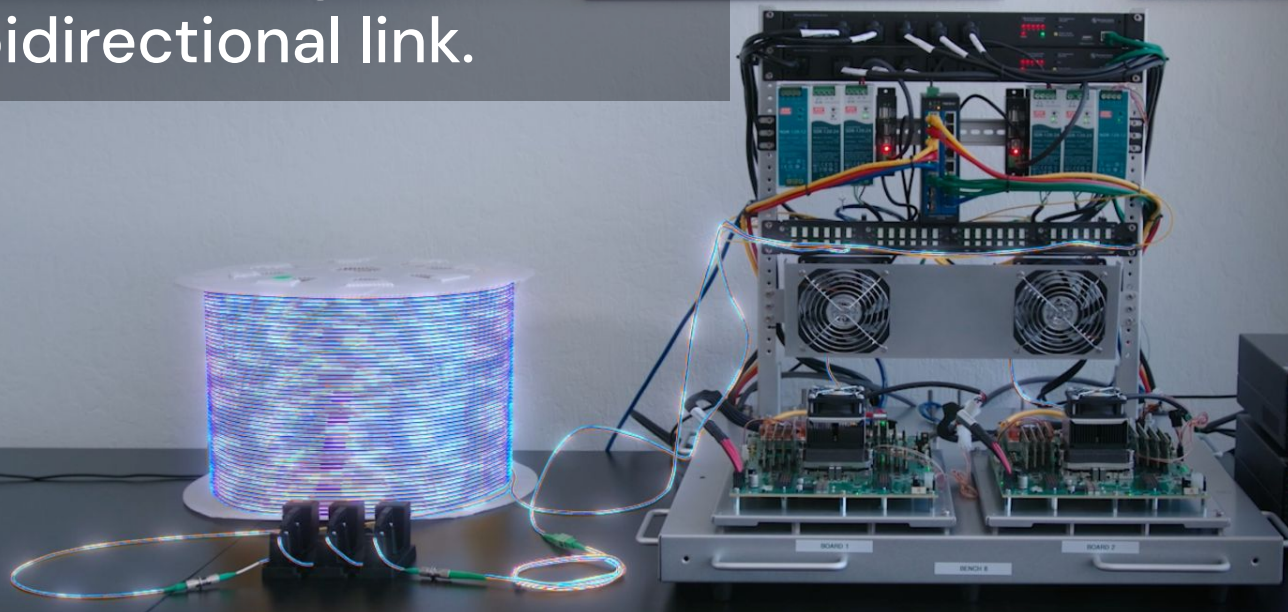


We've built the future.

The future of computing and networking is 3D photonics and big chip packages.

LM builds the fastest chips in the world (Passage) and the light engines that power them (Guide).

World's first 16-wavelength bidirectional link.



- 16 λ
- 800 GBPS / FIBER
- BER < 1E-10, NO FEC
- 2.6 PJ/BIT PHOTONICS
- XSR SERDES
- 1 KILOMETER FIBER
- 6 CONNECTORS
- POLARIZATION INSENSITIVE
- EXTREME TEMP STABILITY
(1200°C/s)



More Details - SerDes Optimization Areas

- Tx Driver Output Swing Control
- RX TIA AGC
- FEC Choice
- Equalization Chain Optimization (Tx and Rx, Linear and Nonlinear)
- Equalization Adaptive Algo Optimization
- Add manual override and monitoring for:
 - Squelch Enable/Disable
 - Signal Detect (SD)
 - Data Valid status flags
- Advanced
 - Slicer thresholds moving independently (asymmetric eyes)
 - CDR Lock with an intentional phase offset.
 - Multipath interference Cancellation